

Recurrent Hypoglycemia After Increased Activity

Andrew · Type 1 · Omnipod + Loop (closed-loop) · Dexcom G7 · Review period: Jun 24 - Aug 23, 2026

Generated from Nightscout CGM & pump data by a family member. Observational — for clinical review, not a diagnosis.

Summary

Marching band began ~Aug 10, adding activity *and* early morning starts. The risk is concentrated in **days he wakes early (~6:30am)**: those days run **4.0% below 54 mg/dL vs 0.9% on late-start days** (target <1%). Lows do **not** occur during activity — they appear ~3–6 h after the first meal (mid-morning) and again ~18–21 h later (overnight). The largest cluster, 10:00–12:00, coincides with a **basal segment of 1.80 U/hr from 09:30** — the highest daytime rate, carried over from an earlier schedule and previously slept through. Bolus insulin has already self-corrected down ~22%; **the basal schedule is unchanged**. He moved into a college dorm Aug 21 and manages lows unsupervised (alarms are heard, fast carbs and a roommate in place; all recorded episodes were self-treated and resolved).

1 · TIME BELOW RANGE, BY WEEK

WEEK	MEAN	TIR 70-180	% <70	% <54 (goal <1%)
July (reference, pre-band)	110	93%	5.1%	0.8%
Aug 3 – 9	113	89%	7.5%	0.8%
Aug 10 – 16	BAND STARTS	115	88%	3.1%
Aug 17 – 23	DORM AUG 21	110	90%	1.8%

Worst individual days: Aug 12 — 14.2% of the day <54; Aug 22 — nadir 40 mg/dL; Aug 23 — nadir 50 mg/dL. Overall glucose and A1c-equivalent remain good (mean ~110, GMI ~5.9%); the problem is hypoglycemia exposure, not average control.

2 · THE DISCRIMINATOR: EARLY-START DAYS

DAY TYPE	DAYS	MEAN	% <70	% <54
Late start (first meal ≥9am)	52	110	6.2	0.9
Early start (first meal <9am)	7	111	12.3	4.0
TIMING ON EARLY DAYS			%<70	%<54
0–3 h after first meal (activity)			4.0	0.0
3–6 h after			25.9	16.3
6–9 h after			12.7	0.0
18–21 h after (overnight)			15.9	10.3

Mean glucose is identical on both day types (110 vs 111) — this is purely a hypoglycemia-exposure difference. Note lows are absent during the activity itself and appear several hours later, matching the family's observation.

3 · EARLY DAYS BY CLOCK HOUR, VS SCHEDULED BASAL

HOUR	MEAN	%<54	BASAL U/HR
01:00	95	13.1	1.45
02:00	104	17.9	1.30
09:00	140	0.0	1.35
10:00	111	14.5	1.80
11:00	93	17.9	1.25
12:00	98	16.7	1.25
17:00	89	8.3	1.70

The 10:00–12:00 cluster begins exactly at the **09:30 step up to 1.80 U/hr** — the highest daytime rate in the schedule, a carry-over from an earlier routine. It was previously slept through; on early days he is awake and active during it.

4 · INSULIN: BOLUS HAS FALLEN, BASAL HAS NOT

	BOLUS INSULIN	CARBS	SCHEDULED BASAL
Before Aug 10	43.3 U/day	178 g/day	38.1 U/day
After Aug 10	33.9 U/day (–22%)	150 g/day	38.1 U/day (unchanged)

Loop and the user have already compensated on the bolus side. The scheduled basal was last tuned Jun 20 for a low-activity summer at home and has not been revisited since activity increased — so the residual excess is concentrated in basal, which is precisely the insulin acting during and after practice.

5 · REPRESENTATIVE NIGHT — SAT AUG 22 (DORM)

00:30	88	→ "Eating Soon" override + 2.5 U
01:00	127	→ "Rage" override (1.2× insulin, target 80) + 2.85 U + 10 g
01:30	76	
02:00	50	
02:30	43	→ treated, 15 g + "Running Low" override
03:00	76	recovered

Post-midnight correction stacking on top of an already-heightened sensitivity state, followed by a crash ~90 minutes later. The overnight "🔥 Rage" preset (1.2× insulin, 80 target) was used 1–8 times per day across this period.

6 · CURRENT SETTINGS & QUESTIONS

Basal (38.05 U/day): 00:00 1.45 · 02:00 1.30 · 04:30 1.35 · 09:30 1.80 · 11:00 1.25 · 16:00 1.80 · 17:00 1.70 · 20:00 2.40 · 22:30 2.20
ISF 53 mg/dL/U · **Carb ratio** 6 g/U · **Target** 100 mg/dL · **DIA** 6 h · flat all day

- The 09:30 → 1.80 U/hr segment.** Highest daytime rate, carried over from an earlier routine, and the 10:00–12:00 severe-low cluster starts on it. Should it come down (or the whole 09:30–16:00 block be flattened toward ~1.25–1.45)?
- An early-start / activity-day protocol** rather than a fixed clock change — his wake time varies from 6:30am to 2pm. Existing overrides are 0.8× / target 120 and 0.7× / target 120. Should one be applied on early days, extended to cover the overnight tail ~18–21 h later?
- Overall basal reduction?** Bolus has already dropped 22% while basal is unchanged from a pre-activity baseline.
- Guidance on late-evening corrections**, given the post-midnight stacking pattern above.
- ISF / carb ratio** — both flat all day and unchanged; should they move with the new activity level?